ZEWEN YANG

Ph.D. Candidate in Geotechnical Engineering

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RESEARCH PROFILE

Ph.D. candidate specializing in outburst-flood hydrodynamics, sediment transport, fluvial erosion and deposition, and landscape evolution. My research combines numerical modelling, flume experiments, field investigations, and theoretical analysis to understand extreme floods on Earth and Mars.

EDUCATION

Sep 2021 - Present

Ph.D. Candidate in Geotechnical Engineering
University of Chinese Academy of Sciences

- Institute of Mountain Hazards and Environment, Chinese Academy of Sciences
- Ph.D. Supervisor: Prof. Weiming Liu

Sep 2017 - Jun 2021

B.Eng. in Water Conservancy and Hydropower Engineering
Yunnan Agricultural University

- Undergraduate Supervisor: Prof. Qiu Yong

VISITING RESEARCH EXPERIENCE

Jan 2026 - Jan 2027

Visiting Ph.D. Student
Geosciences Barcelona (Geo3BCN-CSIC), Barcelona, Spain

- Host Supervisor: Prof. Daniel Garcia-Castellanos
- Joint research on outburst-flood hydrodynamics, sediment transport, and geomorphic evolution.

RESEARCH INTERESTS AND TECHNICAL SKILLS

- Outburst-flood hydrodynamics and geomorphic impacts; landslide-dam breaching and palaeoflood reconstruction
- Erosion, sediment transport, and deposition under unsteady flow; terrestrial and Martian landscape evolution
- Hydrodynamic, sediment-transport, and morphodynamic modelling; flume experiments and field investigations
- Programming: Python and MATLAB

PUBLICATIONS

1. Tang Daobin, Cui Peng, Liu Weiming, Wang Jiao, Yang Zewen, Wang Zili, Yang Anna, and Liu Xiangjun. Flood driving the migration of a large landslide dam and its geomorphic effects. Quaternary Science Reviews, 2026. DOI: 10.1016/j.quascirev.2026.110022.

2. Yang Zewen, Carling Paul A., Liu Weiming, Zhou Gordon G. D., Wang Hao, and Yang Anna. Saturated, Smashed, and Suppressed: The Fate of Suspended Gravel in Large Palaeofloods. Journal of Geophysical Research: Earth Surface, 2026. DOI: 10.1029/2025JF008632.

3. Zhou Yanlian, Liu Weiming, Zhou Liqin, and Yang Zewen. Distribution characteristics of landslide-dammed rivers in the upper Dadu and Min River mountain regions on the eastern Tibetan Plateau. Acta Geographica Sinica, 2025. DOI: 10.11821/dlxb202505010.

4. Yang Zewen, Wu Bingbing, Liu Weiming, Yang Anna, Li Xuemei, Wang Hao, and Zhou Yanlian. Research progress on erosion mechanisms and geomorphic effects of high-energy outburst floods. Earth Science, 2025. DOI: 10.3799/dqkx.2024.009.

5. Yang Anna, Liu Weiming, Wang Hao, Hu Kaiheng, Li Xuemei, Cong Lu, Zhou Yanlian, and Yang Zewen. Dammed lake chronology in the middle Yarlung Tsangpo River: Tracing the origin of late Holocene megafloods. Quaternary Science Reviews, 2025. DOI: 10.1016/j.quascirev.2024.109155.

6. Uwizeyimana David, Liu Weiming, Huang Yu, Habumugisha Jules Maurice, Zhou Yanlian, and Yang Zewen. Spatial distribution characteristics of climate-induced landslides in the Eastern Himalayas. Journal of Mountain Science, 2024. DOI: 10.1007/s11629-024-8869-4.

7. Zhou Yanlian, Liu Weiming, Yanites Brian J., Zhou Liqin, Li Xuemei, and Yang Zewen. Transient geomorphic response after landslide-induced river damming in the eastern margin of the Tibetan Plateau. Catena, 2024. DOI: 10.1016/j.catena.2024.108382.

8. Ruan Hechun, Chen Huayong, Chen Xiaoqing, Zhao Wanyu, Chen Jiangang, Wang Tao, Li Xiao, and Yang Zewen. An investigation of discharge control in landslide dam failures utilizing flexible protecting nets. Engineering Failure Analysis, 2024. DOI: 10.1016/j.engfailanal.2024.108134.

9. Hassan Wajid, Su Feng-huan, Liu Wei-ming, Hassan Javed, Hassan Muzammil, Bazai Nazir Ahmed, Wang Hao, Yang Ze-wen, Ali Muzaffar, and Garcia-Castellanos Daniel. Impact of glacier changes and permafrost distribution on debris flows in Badswat and Shishkat catchments, Northern Pakistan. Journal of Mountain Science, 2023. DOI: 10.1007/s11629-023-7894-5.

10. Wang Hao, Cui Peng, Yang Anna, Tang Jinbo, Wen Shusong, Yang Zewen, Zhou Liqin, Liu Weiming, and Bazai Nazir Ahmed. New evidence of high-magnitude Holocene floods in the Purlung Tsangpo River, southeastern Tibetan Plateau. Catena, 2023. DOI: 10.1016/j.catena.2023.107516.

11. Yang Zewen, Liu Weiming, Garcia-Castellanos Daniel, Ruan Hechun, Luo Junpeng, Zhou Yanlian, and Sang Yunyun. Geomorphic response of outburst floods: Insight from numerical simulations and observations-The 2018 Baige outburst flood in the upper Yangtze River. Science of The Total Environment, 2022. DOI: 10.1016/j.scitotenv.2022.158378.

12. Qiu Yong, Yang Zewen, Zhou Xinyu, Wu Jingang, and Xie Hongying. Prediction of the discharge capacity of right-angle labyrinth weirs using a BP neural network. Water Resources and Power, 2021.

13. Tang Zhifei, Qiu Yong, Yang Zewen, Wang Wenbing, and Yang Kun. Design application of a drop-expansion stilling basin based on a BP neural network. Agricultural Engineering, 2020.

14. Wang Shangjin, Qiu Yong, Yang Zewen, Jiao Xuan, and Zhou Xinyu. Fitting analysis of labyrinth-weir discharge capacity based on variations in overflow-front length. Pearl River, 2020. DOI: 10.3969/j.issn.1001-9235.2020.07.015.

15. Qiu Yong, Lu Huaicha, Zhou Xinyu, Yang Zewen, and Chen Yubin. Effects of side-weir position on the discharge of a right-angle labyrinth weir in an ecological channel. Journal of Zhejiang University of Water Resources and Electric Power, 2020. DOI: 10.3969/j.issn.2095-7092.2020.01.003.

16. Yang Zewen, Qiu Yong, Li Qingmei, Lu Huaicha, and Liu Jian. Influence of sidewall expansion width on stilling-basin length in a drop-type hydraulic-jump stilling basin. Water Resources Planning and Design, 2020.

17. Jiao Xuan, Qiu Yong, Zhang Huaqun, Yang Zewen, and Duan Shengyu. Design of calculation software for energy dissipation by continuous trajectory buckets under low-flow conditions. Water Resources and Power, 2019.

18. Ruan Hechun, Qiu Yong, Luo Guoli, Tang Zhifei, and Yang Zewen. Effects of drop depth on stilling-basin length in an abruptly expanded hydraulic-jump stilling basin. Water Resources Planning and Design, 2019. DOI: 10.3969/j.issn.1672-2469.2019.08.027.

PATENTS

1. Yang Zewen, Liu Weiming, Wang Hao, Yang Anna, Huang Yu, Zhou Yanlian, Zhang Haipeng, and Ruan Hechun. An experimental apparatus and method for efficiently simulating the morphodynamics of outburst floods. Invention patent, under substantive examination.

2. Liu Weiming, Wang Hao, Yang Anna, and Yang Zewen. A method for reconstructing ancient landslide-dam events using a sedimentary-geomorphic evidence chain. Chinese invention patent CN113255124B, granted, 18 November 2022.

3. Qiu Yong, Yang Zewen, Yang Ruiguo, Jiao Xuan, Lu Huaicha, Li Qingmei, Liu Jian, Wang Shangjin, and Zhang Huaqun. A magnetic stirring device for water-sediment separation. Chinese utility model patent CN211097937U, granted, 28 July 2020.

CONFERENCE PRESENTATIONS

• 4th International Conference on Water Disaster Control and Water Environment Regulation, Chengdu, China, November 2023. Participant.

• National Symposium on Geography and Quaternary Research, Taiyuan, China, August 2023. Oral presentation: The 2018 Baige outburst flood and its geomorphic response.

• 6th National Young Geoscientists Conference, Wuhan, China, June 2023. Participant.

• 3rd National Geoscience Graduate Forum, Changchun, China, November 2022. Oral presentation: Research progress on the erosion mechanisms and geomorphic effects of high-energy outburst floods; Best Presentation Award.

• Annual Meeting of the Chinese Hydraulic Engineering Society, online, November 2022. Participant.

• Annual Meeting of the Chinese Hydraulic Engineering Society, Beijing, China, October 2021. Participant.

HONORS AND AWARDS

2025	Overseas Study Scholarship, China Scholarship Council
2024	Outstanding Student Leader, University of Chinese Academy of Sciences
2024	Merit Student, University of Chinese Academy of Sciences
2023	Merit Student, University of Chinese Academy of Sciences
2022	National Scholarship for Graduate Students
2022	Best Presentation Award, 3rd National Geoscience Graduate Forum
2022	Merit Student, University of Chinese Academy of Sciences
2022	First-Class Academic Scholarship, University of Chinese Academy of Sciences
2020	National Scholarship for Undergraduate Students
2019	National Scholarship for Undergraduate Students

ACADEMIC SERVICE

• Secretary, Graduate Student Party Branch (Class of 2023), Institute of Mountain Hazards and Environment, Chinese Academy of Sciences, 2023-2026.

• Class Representative and Student Well-being Liaison, Ph.D. cohort, 2023-2026.

• Reviewer for Journal of Geophysical Research: Earth Surface, Scientific Reports, Frontiers in Earth Science, and Discover Civil Engineering.

LANGUAGES

Chinese (native); English (professional working proficiency)